

HE2003 Econometrics I

Tutorial 5 (Week 6) – Problems & Solutions

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Overview

- **Lecture coverage:** Lecture 5 — *unbiasedness and efficiency of $\bar{Y}$, the Gauss–Markov theorem for the OLS estimators, and the general condition for a linear estimator $\sum a_i Y_i$ to be unbiased*; plus Lecture 4’s decomposition $TSS = ESS + SSR$, used in Q2 to show $R^2 \leq 1$.
- **Questions:** 3 questions from Tutorial Question Set No. 5.
- **Notation:** as in lectures — u_i population error, $\hat{u}_i$ OLS residual; ESS explained, SSR residual, $TSS = ESS + SSR$; “Least Squares Assumption 1/2/3” for the OLS assumptions (never “SLR.1–5”); critical value 1.96 for a two-sided 5% test.

Question 1 (Unbiasedness, efficiency, Gauss–Markov, and BLUE)

Problem

- Explain the notion of unbiasedness and efficiency by using the sample average $\bar{Y}$ as an example.
- Explain the Gauss–Markov Theorem for the OLS estimators.
- Find out the condition that guarantees the linear estimator $\sum_{i=1}^n a_i Y_i$ is an unbiased estimator of the population mean μ .
- Explain the example regarding the efficiency of $\bar{Y}$ on slides 6–8 of Lecture 5.

(Tutorial Question Set No. 5, Q1.)

Solution

Part (a): Unbiasedness and efficiency, via $\bar{Y}$

- Unbiased: $E[\hat{\theta}] = \theta$ – the sampling distribution is centred on the truth.
- Efficient: among unbiased estimators, the smaller Var is more efficient.

Let Y_i be i.i.d. with $E[Y_i] = \mu$, $\text{Var}(Y_i) = \sigma^2$; compare $\bar{Y}$ with $\tilde{Y} = \frac{1}{2}(Y_1 + Y_n)$.

$$E[\bar{Y}] = \frac{1}{n} \sum_{i=1}^n E[Y_i] = \mu \quad (\text{linearity})$$

$$E[\tilde{Y}] = \frac{1}{2}(\mu + \mu) = \mu \quad (\text{linearity})$$

$$\text{Var}[\bar{Y}] = \frac{1}{n^2} \sum_{i=1}^n \text{Var}(Y_i) = \frac{\sigma^2}{n} \quad (\text{independence})$$

$$\text{Var}[\tilde{Y}] = \frac{1}{4}(\sigma^2 + \sigma^2) = \frac{\sigma^2}{2} \quad (\text{Cov}(Y_1, Y_n) = 0)$$

$$\boxed{\text{Var}(\bar{Y}) = \frac{\sigma^2}{n} < \text{Var}(\tilde{Y}) = \frac{\sigma^2}{2} \quad \text{for } n > 2,}$$

Adding Y_1 (unbiased, $\text{Var}(Y_1) = \sigma^2$):

$$\boxed{\text{Var}(\bar{Y}) = \frac{\sigma^2}{n} < \text{Var}(\tilde{Y}) = \frac{\sigma^2}{2} < \text{Var}(Y_1) = \sigma^2.}$$

Part (b): Gauss–Markov Theorem for the OLS estimators

- Assumptions: Least Squares Assumptions 1–3 (LSA1–3: $E[u_i | X_i] = 0$; (X_i, Y_i) i.i.d.; finite fourth moments) and homoskedasticity, $\text{Var}(u_i | X_i) = \sigma_u^2$.
- Result: for any linear unbiased $\tilde{\beta}_0, \tilde{\beta}_1$, $\text{Var}[\hat{\beta}_0] \leq \text{Var}[\tilde{\beta}_0]$ and $\text{Var}[\hat{\beta}_1] \leq \text{Var}[\tilde{\beta}_1]$, i.e. OLS is BLUE.

- Scope: best only among linear unbiased estimators (a biased one can have smaller MSE); no normality needed; without homoskedasticity OLS stays unbiased but is not BLUE (GLS does better).

Part (c): unbiasedness condition for $\sum a_i Y_i$

Method 1: Direct expectation

Let $E[Y_i] = \mu$ for all i , a_i constants, $\hat{\mu} = \sum_{i=1}^n a_i Y_i$.

$$E[\hat{\mu}] = \sum_{i=1}^n a_i E[Y_i] = \mu \sum_{i=1}^n a_i \quad (\text{linearity})$$

$$E[\hat{\mu}] = \mu \quad \forall \mu \iff \left(\sum_{i=1}^n a_i - 1 \right) \mu = 0 \quad \forall \mu \iff \sum_{i=1}^n a_i = 1$$

$$\boxed{\hat{\mu} = \sum_{i=1}^n a_i Y_i \text{ is unbiased for } \mu \iff \sum_{i=1}^n a_i = 1.}$$

Variance (Y_i i.i.d., $\text{Var}(Y_i) = \sigma^2$): $\text{Var}(\hat{\mu}) = \sigma^2 \sum_{i=1}^n a_i^2$. Unbiased: $\sum_{i=1}^n a_i = 1$. Put $a_i = \frac{1}{n} + b_i$, so $\sum_{i=1}^n b_i = 0$:

$$\sum_{i=1}^n a_i^2 = \sum_{i=1}^n \left(\frac{1}{n} + b_i \right)^2 = \frac{1}{n} + \underbrace{\frac{2}{n} \sum_{i=1}^n b_i}_{=0} + \sum_{i=1}^n b_i^2$$

$$\boxed{\sum_{i=1}^n a_i^2 = \sum_{i=1}^n \left(a_i - \frac{1}{n} \right)^2 + \frac{1}{n} \geq \frac{1}{n},}$$

equality $\iff a_i = \frac{1}{n} \forall i$. $\therefore \hat{\mu} = \bar{Y}$ minimises $\text{Var}(\hat{\mu})$, with $\text{Var}(\bar{Y}) = \sigma^2/n$:

$\bar{Y}$ is the Best Linear Unbiased Estimator (BLUE) of μ .

Method 2: Cauchy–Schwarz

$$\sum_{i=1}^n a_i = 1 \implies 1 = \left(\sum_{i=1}^n a_i \cdot 1 \right)^2 \leq \left(\sum_{i=1}^n a_i^2 \right) \left(\sum_{i=1}^n 1^2 \right) = n \sum_{i=1}^n a_i^2 \quad (\text{C-S})$$

$$\implies \text{Var}(\hat{\mu}) = \sigma^2 \sum_{i=1}^n a_i^2 \geq \frac{\sigma^2}{n} = \text{Var}(\bar{Y})$$

Equality in C-S $\iff a_i$ constant $\iff a_i = \frac{1}{n} \forall i$. $\therefore \bar{Y}$ is BLUE.

Part (d): the Lecture 5 slides 6–8 simulation

- 20 samples of size $n = 10$ from $N(2, 1)$: means of Y_1 and $\bar{Y}$ are about 1.89 and $1.96 \approx \mu = 2$ – both unbiased.
- $\bar{Y}$ ranges over $[1.16, 2.58]$, Y_1 over $[-0.64, 4.27]$; $\bar{Y}$ is closer to 2 in 16 of 20 samples – $\bar{Y}$ is more efficient.
- $\text{Var}(Y_1) = \sigma^2 = 1$, $\text{Var}(\bar{Y}) = \sigma^2/n = 0.1$ (10 times smaller), matching the spread.

$\text{Var}(Y_1) = 1, \quad \text{Var}(\bar{Y}) = 0.1; \quad \bar{Y} \text{ is unbiased and far more efficient than } Y_1.$

Question 2 (Why R^2 cannot exceed 1)

Problem

In this question we show why R^2 cannot be larger than 1.

- (a) Show that $TSS = \sum_{i=1}^n [\hat{u}_i + (\hat{Y}_i - \bar{Y})]^2$.
- (b) Show that $TSS = ESS + SSR$, where TSS and ESS are as defined in the lecture slides and $SSR = \sum_{i=1}^n \hat{u}_i^2$. [Hint: use $\sum_{i=1}^n \hat{u}_i \hat{Y}_i = 0$ and $\sum_{i=1}^n \hat{u}_i = 0$.]
- (c) Show that $R^2 \leq 1$.

(Tutorial Question Set No. 5, Q2.)

Solution

Method 1: Algebraic decomposition

Part (a).

$$Y_i - \bar{Y} = \hat{u}_i + (\hat{Y}_i - \bar{Y}) \quad (Y_i = \hat{Y}_i + \hat{u}_i)$$

$$\implies TSS = \sum_{i=1}^n (Y_i - \bar{Y})^2 = \sum_{i=1}^n [\hat{u}_i + (\hat{Y}_i - \bar{Y})]^2 \quad \blacksquare$$

Part (b).

$$\begin{aligned} TSS &= \sum_{i=1}^n \hat{u}_i^2 + \sum_{i=1}^n (\hat{Y}_i - \bar{Y})^2 + 2 \sum_{i=1}^n \hat{u}_i (\hat{Y}_i - \bar{Y}) \quad (\text{expand (a)}) \\ &= SSR + ESS + 2 \left[\sum_{i=1}^n \hat{u}_i \hat{Y}_i - \bar{Y} \sum_{i=1}^n \hat{u}_i \right] \\ \sum_{i=1}^n \hat{u}_i \hat{Y}_i &= \hat{\beta}_0 \sum_{i=1}^n \hat{u}_i + \hat{\beta}_1 \sum_{i=1}^n \hat{u}_i X_i = 0 \quad (\text{FOCs: } \sum \hat{u}_i = 0 = \sum \hat{u}_i X_i) \\ \sum_{i=1}^n \hat{u}_i &= 0 \quad (\text{FOC}) \\ \implies TSS &= SSR + ESS \end{aligned}$$

$$\boxed{TSS = SSR + ESS = ESS + SSR.} \quad \blacksquare$$

Part (c).

$$0 \leq SSR = TSS - ESS \leq TSS \quad (SSR = \sum_{i=1}^n \hat{u}_i^2 \geq 0, ESS \geq 0)$$

$$\implies R^2 = \frac{ESS}{TSS} = 1 - \frac{SSR}{TSS} \in [0, 1]$$

$$\boxed{0 \leq R^2 = 1 - \frac{SSR}{TSS} \leq 1.} \quad \blacksquare$$

Method 2: orthogonality of residuals

$$Y - \bar{Y}\mathbf{1} = \hat{u} + v, \quad v_i := \hat{Y}_i - \bar{Y}$$

$$\hat{u} \cdot v = \sum_{i=1}^n \hat{u}_i(\hat{Y}_i - \bar{Y}) = 0 \quad (\text{FOCs})$$

$$\implies \text{TSS} = \|\hat{u} + v\|^2 = \|\hat{u}\|^2 + \|v\|^2 = \text{SSR} + \text{ESS} \quad (\text{Pythagoras})$$

$$\implies \text{ESS} \leq \text{TSS} \iff R^2 \leq 1 \quad (\text{SSR} = \|\hat{u}\|^2 \geq 0)$$

Final answer

$$\boxed{\text{TSS} = \sum_{i=1}^n [\hat{u}_i + (\hat{Y}_i - \bar{Y})]^2 = \text{ESS} + \text{SSR}, \quad 0 \leq R^2 = \frac{\text{ESS}}{\text{TSS}} = 1 - \frac{\text{SSR}}{\text{TSS}} \leq 1.}$$

Question 3 (Comparing weekly working hours: SG versus HK)

Problem

Suppose that the population mean and variance of employees' working hours per week in Singapore (SG) are μ_1 and σ_1^2 , respectively, and N_1 i.i.d. sample observations were drawn from this population; the population mean and variance of employees' working hours per week in Hong Kong (HK) are μ_2 and σ_2^2 , respectively, and N_2 i.i.d. sample observations were drawn from this population.

The SG and HK random samples are totally independent. We are interested in $\mu_1 - \mu_2$, the difference between the two population means, and use $\hat{d} = \bar{Y}_1 - \bar{Y}_2$ as its estimator, where $\bar{Y}_1$ is the sample average of the N_1 SG observations and $\bar{Y}_2$ is the sample average of the N_2 HK observations.

**The formulas of means, variances, and covariances of sums of random variables can be found in Key Concept 2.3 (p.74) of "Introduction to Econometrics" or in the Supplement "Expectation, Variance, and Covariance".*

(a) Compute $E[\bar{Y}_1]$, $E[\bar{Y}_2]$, $\text{Var}[\bar{Y}_1]$, and $\text{Var}[\bar{Y}_2]$.

(b) Compute $E[\hat{d}]$, $\text{Cov}[\bar{Y}_1, \bar{Y}_2]$, and $\text{Var}[\hat{d}]$.

(Tutorial Question Set No. 5, Q3.)

Solution

Method 1: Direct expectation/variance algebra

Part (a). SG sample $Y_{1,i}$ i.i.d., $E[Y_{1,i}] = \mu_1$, $\text{Var}(Y_{1,i}) = \sigma_1^2$; HK identical with index 2.

$$E[\bar{Y}_1] = \frac{1}{N_1} \sum_{i=1}^{N_1} E[Y_{1,i}] = \mu_1 \quad (\text{linearity})$$

$$\text{Var}[\bar{Y}_1] = \frac{1}{N_1^2} \sum_{i=1}^{N_1} \text{Var}(Y_{1,i}) = \frac{\sigma_1^2}{N_1} \quad (\text{independence})$$

$$\boxed{E[\bar{Y}_1] = \mu_1, \quad E[\bar{Y}_2] = \mu_2, \quad \text{Var}[\bar{Y}_1] = \frac{\sigma_1^2}{N_1}, \quad \text{Var}[\bar{Y}_2] = \frac{\sigma_2^2}{N_2}.}$$

Part (b).

$$E[\hat{d}] = E[\bar{Y}_1] - E[\bar{Y}_2] = \mu_1 - \mu_2 \quad (\text{linearity})$$

$$\text{Cov}[\bar{Y}_1, \bar{Y}_2] = \frac{1}{N_1 N_2} \sum_{i=1}^{N_1} \sum_{j=1}^{N_2} \text{Cov}(Y_{1,i}, Y_{2,j}) = 0 \quad (\text{bilinearity; independent samples})$$

$$\boxed{\text{Cov}[\bar{Y}_1, \bar{Y}_2] = 0.}$$

$$\text{Var}[\hat{d}] = \text{Var}[\bar{Y}_1] + \text{Var}[\bar{Y}_2] - 2 \text{Cov}[\bar{Y}_1, \bar{Y}_2] = \frac{\sigma_1^2}{N_1} + \frac{\sigma_2^2}{N_2}$$

$$\text{Var}[\hat{d}] = \frac{\sigma_1^2}{N_1} + \frac{\sigma_2^2}{N_2}.$$

Standard error and 95% CI. With sample variances s_1^2, s_2^2 and large N_1, N_2 (CLT):

$$\text{CI}_{95\%}(\mu_1 - \mu_2) = (\bar{Y}_1 - \bar{Y}_2) \pm 1.96 \text{SE}(\hat{d}) = (\bar{Y}_1 - \bar{Y}_2) \pm 1.96 \sqrt{\frac{s_1^2}{N_1} + \frac{s_2^2}{N_2}}.$$

Final answer

$$E[\bar{Y}_1] = \mu_1, \quad E[\bar{Y}_2] = \mu_2, \quad \text{Var}[\bar{Y}_1] = \frac{\sigma_1^2}{N_1}, \quad \text{Var}[\bar{Y}_2] = \frac{\sigma_2^2}{N_2}.$$

$$E[\hat{d}] = \mu_1 - \mu_2, \quad \text{Cov}[\bar{Y}_1, \bar{Y}_2] = 0, \quad \text{Var}[\hat{d}] = \frac{\sigma_1^2}{N_1} + \frac{\sigma_2^2}{N_2}.$$

Final Answers

- Q1(a)** $\text{Var}(\bar{Y}) = \sigma^2/n < \text{Var}(\tilde{Y}) = \sigma^2/2 < \text{Var}(Y_1) = \sigma^2$ for $n > 2$; all unbiased.
- Q1(b)** LSA1–3 + homoskedasticity $\Rightarrow \hat{\beta}_0, \hat{\beta}_1$ BLUE (smallest variance among linear unbiased).
- Q1(c)** $\sum_{i=1}^n a_i Y_i$ unbiased $\iff \sum_{i=1}^n a_i = 1$; minimum variance at $a_i = 1/n$, so $\bar{Y}$ is BLUE.
- Q1(d)** $\text{Var}(Y_1) = 1$, $\text{Var}(\bar{Y}) = 0.1$; both unbiased, $\bar{Y}$ far more efficient.
- Q2** $\text{TSS} = \sum [\hat{u}_i + (\hat{Y}_i - \bar{Y})]^2 = \text{ESS} + \text{SSR}$; $0 \leq R^2 = \text{ESS}/\text{TSS} = 1 - \text{SSR}/\text{TSS} \leq 1$.
- Q3** $E[\bar{Y}_1] = \mu_1$, $E[\bar{Y}_2] = \mu_2$, $\text{Var}[\bar{Y}_1] = \sigma_1^2/N_1$, $\text{Var}[\bar{Y}_2] = \sigma_2^2/N_2$; $E[\hat{d}] = \mu_1 - \mu_2$, $\text{Cov}[\bar{Y}_1, \bar{Y}_2] = 0$, $\text{Var}[\hat{d}] = \sigma_1^2/N_1 + \sigma_2^2/N_2$.